



3D printed circuit boards

Next Dynamics builds a new 3D printer that can even build circuit boards using multiple materials. <http://dgit.in/3Dcrtbrd>



Knowmail raises \$3.5M

Knowmail is an Israeli startup that wants to use AI to help better managing of emails and other messages. <http://dgit.in/KnowmailAI>

Bazaar

LG V20

Niche appeal

If you're one of those looking for a removable battery, the LG V20 is the only flagship phone that suits your needs. Despite an all metal build, the phone allows you to pop the back and remove the battery.

That said, the V20's metallic build doesn't feel as premium. The phone's MIL-STD-810G certification comes at the expense of a premium feel. You don't get the anodised aluminium that premium smartphones are often known for. Given that shock proofing doesn't really protect your display in practical scenarios, we would have happily dropped it for a more premium device.

LG has chosen an LCD panel this time, and the results are quite evident. While the V20 is good in terms of colour reproduction and viewing angles, the black levels are noticeably lower. Moreover, there's some light leak near the edges, which is more evident when the primary display is turned off. The part just below the secondary panel will look somewhat greyish in this case.

The secondary display remains more of a gimmick. You can put your most used apps on this, or have the quick settings here, but it doesn't really solve a purpose, or become a feature you couldn't do without. Your phone looks different, but that's about it.

That aside, the V20 is your regular flagship. The Snapdragon 820 is as fast as ever, however, the QHD display does reduce frame rates on gaming a bit. This will show up in benchmarks, but is hard to spot in real life. In general, the LG V20 is as fast as any other flagship out there, but you'll have to take the phone off its power saving mode to draw the full power of the SoC. On the default mode, games lag quite a bit.

The V20 also has a dual-camera setup on the back. Unlike the Huawei P9 or iPhone 7 Plus, though, it doesn't combine the power of two lenses. Like the G5, you get a wide angle 8MP camera and a 16MP rear camera. The image



Price
₹54,999

quality is also just as good, which means it is no match for a Pixel or iPhone right now. The wide angle camera produces fish-eye like images, which some may like, while many will not enjoy.

The LG V20 can take good images under most lighting scenarios, and would have been amongst the best some months ago. It's just that the Pixel and iPhone 7 Plus have changed the game recently, with the former being a major deterrent to the V20. The Pixel has a much higher dynamic range, and better details and colours than this one.

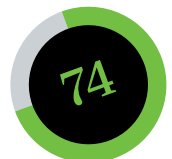
The LG V20 is also behind in terms of battery life. The phone drops 5 per cent battery

over 15 minutes of gaming and runs for just over five hours on the PC Mark battery test. This is satisfactory by industry standards, but one still wishes for more. In general, the V20 will only last a work day at best.

Where the LG V20 truly excels is audio quality, and that's not just in terms of output. The V20 is the best smartphone for those looking for good audio recording quality. A song played on an iPhone SE and recorded via the V20, sounds better on the latter. The phone combines three Acoustic Overload Point microphones to record audio, while a 32-bit Hi-Fi DAC handles audio output. In both cases, the V20 is the best you can get, rivalled only by the HTC 10, and that too in audio output only. The LG V20 is a tad louder than the HTC 10, too.

To conclude, the LG V20 is a very good smartphone, but it's meant for a very niche user. If the removable back and battery, or good audio performance are not your priorities, you really won't make the best purchase with the V20. It is one of the few phones to come with Android Nougat out-of-the-box, but so is the Google Pixel. At a ₹50k+ price point, Google's smartphones can do almost everything better than the V20, barring audio performance.

Prasid Banerjee



Features72

Performance.....70

Value69

Design83

Specifications

Display: 5.7-inch QHD IPS LCD, 2.1-inch 160 x 1040 pixel secondary display; **RAM:** 4 GB; **SoC:** Qualcomm Snapdragon 820; **Operating System:** Android 7.0 Nougat; **Storage:** 32/64 GB; **Battery:** 3200 mAh; **Camera:** 16 MP primary + 8 MP wide angle, 5 MP Front Camera

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